

Cellular Networks (304192) - Quick Revision Notes

1. Hata Model

Used for path loss prediction in urban areas. Depends on frequency, antenna height, and distance.

2. OFDM

Orthogonal Frequency Division Multiplexing divides bandwidth into multiple orthogonal subcarriers.

3. BER

Bit Error Rate measures error probability. Rayleigh channel has higher BER than AWGN.

4. MIMO

Multiple Input Multiple Output improves capacity and reliability using multiple antennas.

5. Link Budget

Received Power = Transmitted Power - Losses. Check feasibility using receiver sensitivity.

Important Formulas

Path Loss, SNR, Capacity (Shannon), BER expressions

Exam Tips

Focus on numericals, definitions, and diagrams for 10-mark questions.